

IAC-13-B2.1.3

GNSS PERFORMANCES FOR MEO, GEO AND HEO

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Global Navigation Satellite Systems (GNSSs) such as GPS, GLONASS, and the future Galileo and BeiDou, have demonstrated to be a valid and efficient system for various space applications in Low Earth Orbit (LEO), such as spacecraft orbit and attitude determination, rendezvous and formation flight of two or more spacecrafts, and timing synchronization. A GNSS presents a number of significant advantages, in particular for small satellites: it provides an autonomous navigation system, which requires just a relatively inexpensive realization and installation cost of the on board GNSS receiver, with low power consumption, limited mass and volume. Nowadays, the GNSS technology for LEO satellites is often used, thanks to the large number of visible satellites, the good geometry coverage and the strong signal power. However, the research of GNSS solutions for Medium Earth Orbit (MEO), Geostationary Earth Orbit (GEO) and High Earth Orbit (HEO) satellites is still new. In this context, this study aims to estimate accurately the GPS and Galileo performances for MEO, GEO and HEO use, such as for lunar applications. Unlike most of the previous investigations, our study is making use of a very accurate multi-GNSS full constellation simulator "Spirent GSS8000", which supports simultaneously the GPS and Galileo systems and the L1, L5, E1, E5 frequency bands. Performances offered by GPS and by GPS-Galileo-combined systems are evaluated in terms of availability, pseudorange error factors, geometry factors, Doppler shifts and Doppler rates.

I. INTRODUCTION

Although primarily conceived as military navigation systems for land, sea and airborne users, nowadays Global Navigation Satellite Systems (GNSSs) such as GPS, GLONASS, and the future Galileo and BeiDou, can also be considered as effective and efficient systems for a considerable number of space applications. These include autonomous real time spacecraft position and attitude determination, precise orbit determination, rendezvous, formation flight of two or more spacecraft and timing synchronization [1]. Moreover, a GNSS seems to be suitable for small satellites, because it requires relatively inexpensive realization and installation cost of the on-board GNSS receiver, which has low power consumption, limited mass and volume, and provides an autonomous navigation system. Projections show that over the next twenty years approximately [2]: 60% of space missions will operate in LEO (which ranges from an altitude of 100 km to about 2000 km), 35% of space missions will operate at higher altitudes up to GEO (altitude of approximately 36 000 km) and the rest will be a Cislunar / Interplanetary or HEO missions. Today, thanks to the large number of visible satellites, the good geometry coverage and the strong signal power, many GNSS receivers are already successfully flying on satellites in LEO orbits. In fact, most LEO space users

share similar operational benefits as more "traditional" Earth users. However, space remains a challenging operational environment at higher altitudes, such as MEO, GEO and HEO, where the GNSS receiver performance and the GNSS solution (e.g. navigation solution) are considerably affected. The receiver performances are in fact strongly influenced by high spacecraft translational and rotational dynamics, weaker received signals power, thermo-mechanical stresses and possible multipath effects, self-induced from the nearby surfaces or due to reflection with other vehicles. Moreover, the GNSS solution may not even exist if a minimum number of GNSS satellites are not in the line of sight (LOS) or of course if the GNSS receiver is not able to acquire their signals. If a GNSS solution exists, i.e. for code-based observations, its error will depend on the product between a geometry factor (the composite effect of the relative satellite-user geometry on the GNSS solution error) and a pseudorange error factor (a statistical sum of the contributions from each of the ranging error sources) [3]. In particular, the higher will be the orbit above the GNSS constellation, the larger will be the geometry error factor and accordingly the larger will be the GNSS solution error.

The aim of this paper is to estimate accurately the GPS and GPS-Galileo-combined performances for MEO, GEO and HEO use, such as for lunar transfer

trajectories, in terms of availability, pseudorange error factors and geometry factors. Furthermore, the Doppler shift and the Doppler rate are calculated for the considered orbital cases, being an influential parameter in the GNSS receiver design. Unlike most of the previous investigations, this study is making use of the very accurate multi-GNSS constellation simulator “Spirent GSS8000”, which supports simultaneously the GPS and Galileo systems and the L1, L5, E1, E5 frequency bands. Spirent simulator includes facilities to accommodate to the special needs of space-based receiver testing, including [4]:

- Full account for the double atmosphere effect of signals passing through the atmosphere twice for the GNSS satellites located on the far side of the Earth.
- Realistic satellite transmit-antenna patterns.
- Spacecraft models and spacecraft motion models.
- Ability to define trajectory data, including in real-time.

II. SIMULATION MODELS AND ASSUMPTIONS

As already mentioned, the GNSS performances are evaluated for the GPS constellation and for the GPS-Galileo-combined constellation. The initial time of the performed simulations is arbitrarily selected as February 13, 2013, 0:00:00 UTC.

II.I Atmosphere Model

The areas of the atmosphere, known as the ionosphere and the troposphere, delay the RF signal from each GNSS satellite to the receiver. In order to calculate the true range from each satellite, and hence the receiver position, these delays must be taken into account [5]. To calculate the tropospheric delay, the simulator uses the tropospheric model from reference [6]. Regarding the ionospheric delay, for Galileo satellites, it uses the NeQuick ionospheric model described in [5] and [7], which applies equally well to both terrestrial and space-borne receivers. For GPS satellites, the ionospheric delay is modelled according to the Klobuchar model [8]. Furthermore, because the Klobuchar model is not applicable at altitudes within, or above, the ionosphere, the simulator switches for altitudes above 80 km between the Klobuchar model and an alternative one, defined in [5], which takes into account the reduction in the ionization level (Total Electron Count, or TEC) with increasing height in the ionospheric layer.

II.II Constellations Model

The constellations model consists of 31 GPS satellites, including at least four satellites in each of six orbital planes, as described in [8], and 27 Galileo satellites as in the “standard” Galileo constellation defined in [9].

II.III Signals Model

In this study just the L1 GPS and E1 Galileo frequencies are considered, for which it is assumed respectively a power reference level of -130 dBm and of -125.5 dBm. Each satellite signal strength is modelled to provide realistic signal levels at the receiver position P_r by using the following formula from [5]:

$$P_r = P_{ICD} + O_G + 20 \times \log_{10} \left(\frac{R_0}{R} \right) - L_{TX} - L_{RX} \quad (\text{dBm}) \quad [1]$$

where :

P_{ICD} is the guaranteed minimum signal level for the GNSS satellite, here assumed -130 dBm for GPS and -125.5 dBm for Galileo.

O_G “Global Offset”. In Spirent simulator, the default value is 10 dB for both constellations [5]. This value is chosen to match the performance obtained when using the simulator with the performance obtained when using a real antenna capturing a real signal under good conditions (i.e., a clear view of the sky). More specifically, part of this offset results from the higher transmit power of the satellites as compared to the minimum signal specifications, and part compensates for the higher thermal noise floor in the simulations than as captured by a real antenna pointing to the sky (i.e., the real antenna temperature is typically much lower than the ambient temperature).

R_0 is the reference range used for inverse-square variation calculation and equal to the range from a receiver to the GNSS satellite at zero elevation.
 $R_0 = \sqrt{(\text{satellite orbit radius})^2 - (\text{earth radius})^2}$

R is the range from GNSS satellite to the receiver.

L_{TX} is the loss from the GNSS satellite transmit antenna in the direction of the receiver.

L_{RX} is the loss from the receiver antenna in the direction of the GNSS satellite.

II.IV Transmitter Antenna Patterns

The transmitter antenna level pattern is modeled to simulate the directional (angular) dependence of the strength of the radio waves from the GNSS transmitter antenna. Since the GNSS transmitter antenna points to the Earth (the antenna main lobe is directed to the Earth, to serve the Earth user), this has a significant effect for

orbiting space vehicles, particularly for those whose orbit is above the GNSS constellation and do not lie on (or close to) the boresight of the GNSS transmitting antenna. At the time of carrying out this study, not enough data were available about the Galileo transmitter antenna pattern side lobes. Therefore, we used the GPS transmitter antenna pattern from Block II-A (as defined in [10] and provided by Spirent) to model all the transmitters for both simulated constellations. This specific antenna pattern is plotted in Figure 1 as a function of the antenna elevation and in Figure 2 as a 3D function of the antenna elevation and azimuth.

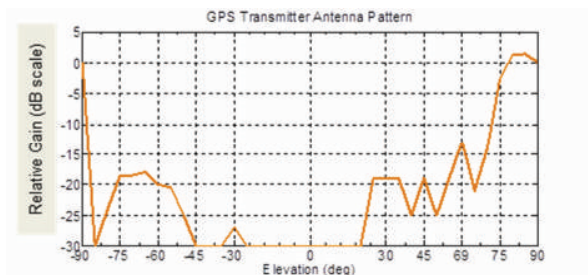


Fig. 1: simulated GPS Transmitter Antenna Pattern (based on [10] for Block II-A). The boresight is at 90°.

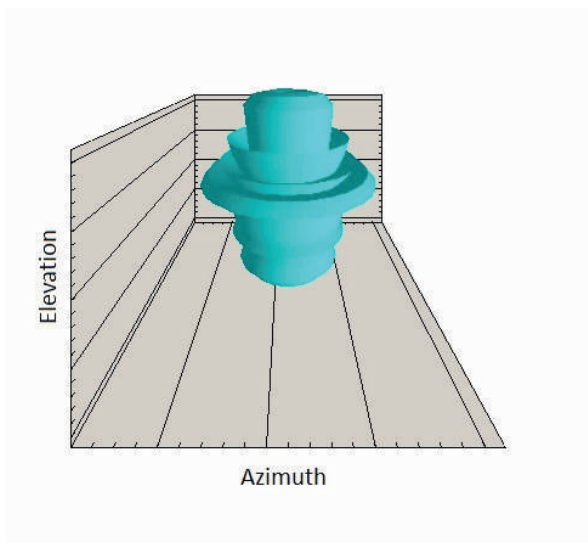


Fig. 2: 3D view of the simulated GPS Transmitter Antenna Pattern (based on [10] for Block II-A).

II.V Receiver Antenna Pattern

To provide an easy and conservative reference, the receiver antenna is assumed to be isotropic (0 dBi).

Receiver Kinematics Models

The following orbital cases are investigated:

- ✈ MEO under GPS constellation.
- ✈ MEO above GPS constellation.
- ✈ GEO.
- ✈ HEO highly elliptical as lunar transfer orbit, from 185 km earth orbit parking to 100 km lunar orbit parking.

The following table contains the considered orbit definitions, while figure 3 provides an illustration of the orbits. For all the orbits Right ascension, Mean anomaly and Argument of perigee are set to be zero at the initial time.

MEO 1 under GPS constellation	
Semi-major axis	16371 km
Eccentricity	0 °
Inclination	0 °
Period	5 _h 47 _{min} 26 _s
Orbital velocity	4.93 km/s

MEO 2 above GPS constellation	
Semi-major axis	31371 km
Eccentricity	0 °
Inclination	0 °
Period	15 _h 21 _{min} 37 _s
Orbital velocity	3.56 km/s

GEO	
Semi-major axis	42168 km
Eccentricity	0 °
Inclination	0 °
Period	23 _h 56 _m 4 _s
Orbital velocity	3.07 km/s

HEO	
Semi-major axis	198613.5 km
eccentricity	0.966991166 °
inclination	5 °
period	10 _{days} 5 _h 20 _{min} 55 _s
perigee	185 km
apogee	390671 km
Max Orbital velocity	10.94 km/s
Min Orbital velocity	0.18 km/s

Table 1: Data of the investigated orbits: MEO 1 (under GPS constellation), MEO 2 (above GPS constellation), GEO and HEO.

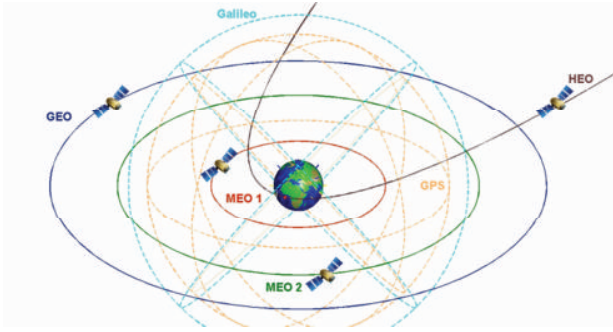


Fig. 3: Matlab plot of the four investigated orbit and of the GPS and Galileo satellites orbits.

The attitude orientation of the space vehicle that hosts the GNSS receiver is set to be inertial (anyway, since the receiver antenna is assumed to be isotropic, the attitude does not affect the signals reception).

II. VI Performances Metrics

The GNSS performances are investigated in order to verify if a stand-alone GNSS solution exists and in the case it exists, how much it can be accurate. The evaluated metrics are:

- *Received power P_r* at the orbiting GNSS receiver (the first four highest power signals are considered at any instant t)
- *Visibility of the GNSS satellite s_i* : it is satisfied if the GNSS satellite s_i is within the receiver line of sight (LOS), that means it is not obstructed by any other object (i.e., the Earth).
- *Availability of the GNSS satellite s_i* . This can be seen as a Boolean variable which is true at the time t only if:
 - The GNSS satellite is visible at t .
 - At t , the received power from the GNSS satellite s_i is higher than a defined threshold $power_{th}$.
- *Time Offset* which affects the pseudorange error factor. The total time offset is due to: receiver clock offset, satellites clock offset, receiver noise and interference, multipath offset, receiver hardware offsets and delays due to the atmosphere [3].
- *Dilution Of Precision (DOP)* which is the effect of the relative satellite/user geometry on the GNSS navigation solution.
- *Doppler shifts and Doppler rates*. The Doppler effect is a change in the apparent frequency of the received signal caused by the relative motion of the emitter and receiver [11].

III. SIMULATION RESULTS

III. I GPS Constellation

The following subsections describe the simulations results obtained by taking into consideration the GPS constellation only. In the simulations the performance metrics are calculated during one full orbital period.

Received Power

Table 2 summarizes the minimum and maximum obtained *received power* levels P_r , as outcomes of the simulations for the four orbital cases. These are reported as ranges of possible received power levels, obtained in one full orbit, for the cases when 1, 2, 3 or 4 GPS satellites need to be acquired or tracked simultaneously. The underlined values, being the minimum value of the range, can be considered as the minimum power threshold ($power_{th}$) for a receiver that should be able to acquire and/or track at least four satellites simultaneously, when travelling in the corresponding orbits. Hence, these $power_{th}$ values, for each orbital cases, represent a necessary minimum condition for the navigation solution existence.

	1 GPS satellite	2 GPS satellites	3 GPS satellites	4 GPS satellites
MEO 1				
$P_{r,max}$ (dBm)	-110.6	-122.6	-122.7	-126.1
$P_{r,min}$ (dBm)	-126.3	-128.6	-134.1	<u>-134.2</u>
MEO 2				
$P_{r,max}$ (dBm)	-124.1	-125.5	-125.6	-129.2
$P_{r,min}$ (dBm)	-135.7	-140.4	-140.7	<u>-141.2</u>
GEO				
$P_{r,max}$ (dBm)	-127.0	-127.1	-127.1	-130.8
$P_{r,min}$ (dBm)	-140.7	-144.5	-144.8	<u>-144.9</u>
HEO				
$P_{r,max}$ (dBm)	-113.6	-115.5	-116.6	-116.9
$P_{r,min}$ (dBm)	-162.6	-162.6	-162.7	<u>-162.7</u>

Table 2: *Received power* levels ranges, for the four orbital cases. The underlined minimum power $P_{r,min}$ of the ranges can be considered as the minimum power threshold $P_{r,th}$ to capture four GPS satellites simultaneously.

Figures 4, 5, 6 and 7 illustrate the probability that a received power is higher than a power threshold, for the four orbital cases, if 1, 2, 3, 4, or 5 GPS satellites have to be acquired and/or tracked simultaneously. They were computed by assuming a normal distribution (verified for all four orbits with 5% significance level) for the received power levels and computing the mean and the standard deviation from the simulation data for each full orbit. We note that in figure 7 the curves slope

is lower than in the other cases. This is due to the highly elliptic shape of the orbit, which makes the range of possible received power levels and thus the power level dispersion larger.

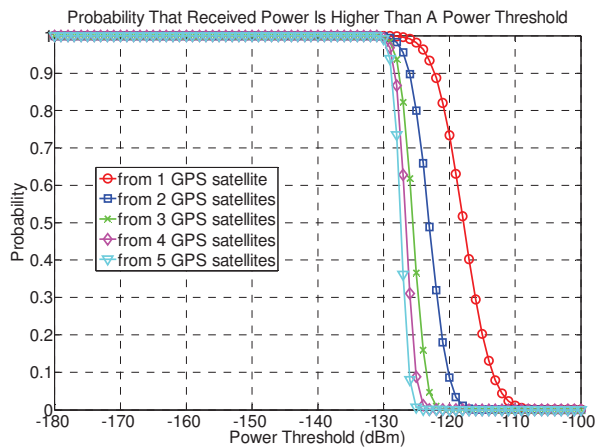


Fig. 4: Probability that received power in MEO 1 is higher than a given power threshold for 1, 2, 3, 4, or 5 GPS satellites simultaneously.

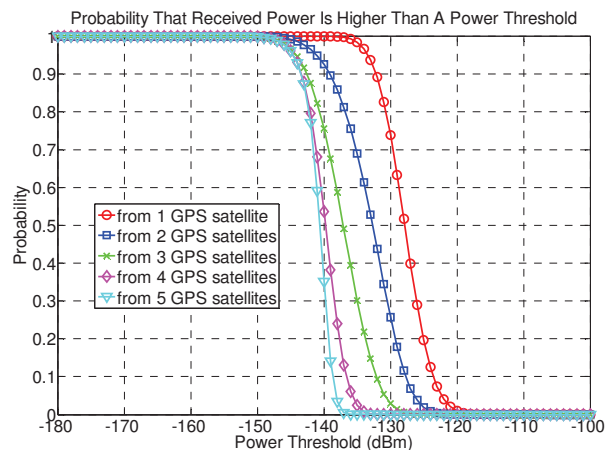


Fig. 5: Probability that received power in MEO 2 is higher than a given power threshold for 1, 2, 3, 4, or 5 GPS satellites simultaneously.

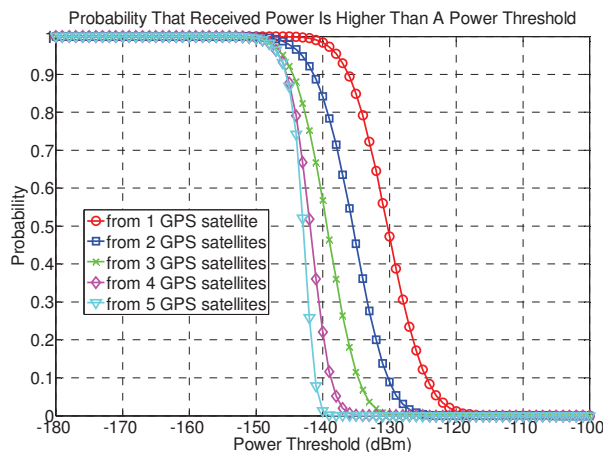


Fig. 6: Probability that received power in GEO is higher than a given power threshold for 1, 2, 3, 4, or 5 GPS satellites simultaneously.

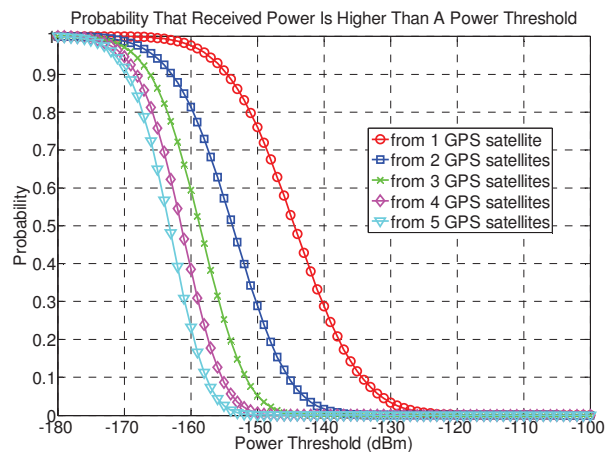


Fig. 7: Probability that received power in HEO is higher than a given power threshold for 1, 2, 3, 4, or 5 GPS satellites simultaneously.

Moreover, the normal fit is not as tight for the HEO case as for the other cases. Indeed, if we consider the power levels range obtained in one full orbit simulation, for four GPS satellites, the minimum value experienced is -162.7 dBm, but if we refer to figure 7 and to Table 5, as we can read, there is only 95% of probability that the power level is higher than -170 dBm, 5% of the times the power level could be even lower. Hence, there is not

a perfect fitting between the normal probability distribution and the limited series of data obtained in one orbit simulation. This deviation is considerable in the HEO case because, due to highly elliptical trajectory, the receiver experiences the signals characteristics corresponding to many different altitudes and just a change only in the initial time of the

performed simulation would change the power levels range.

Visibility and Availability

The *visibility* is a necessary condition for the *availability* and then it is implicitly satisfied when also the availability is satisfied.

Although a theoretical lower bound can be defined, the power threshold $power_{th}$ cannot be chosen uniquely, because it should depend on the specific receiver design considered. In order to not limit the investigation on just one power threshold $power_{th}$, the availability is calculated for different power thresholds $power_{th}$ equal to -160, -140, -120, -115 dBm. The figures 8, 9, 10 and 11 show the GPS satellites availability time trend for

those power thresholds, for each GPS satellite, for each orbital case. These figures indicate visually how long each satellite is available without outages and how often they are available in one whole orbit. In particular, as expected, in figure 11 it is clear that most of the available satellites are concentrated at the beginning and at the end of the orbital period, which are the times when the receiver is at the perigee. The same information are reported quantitatively in table 6 that is described in details in section III.II.

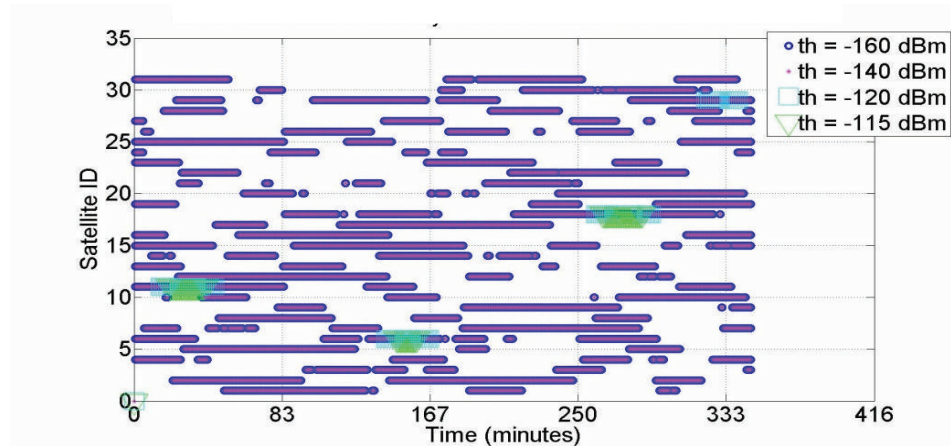


Fig. 8: GPS Satellites Availability for a MEO receiver at 10000 km.

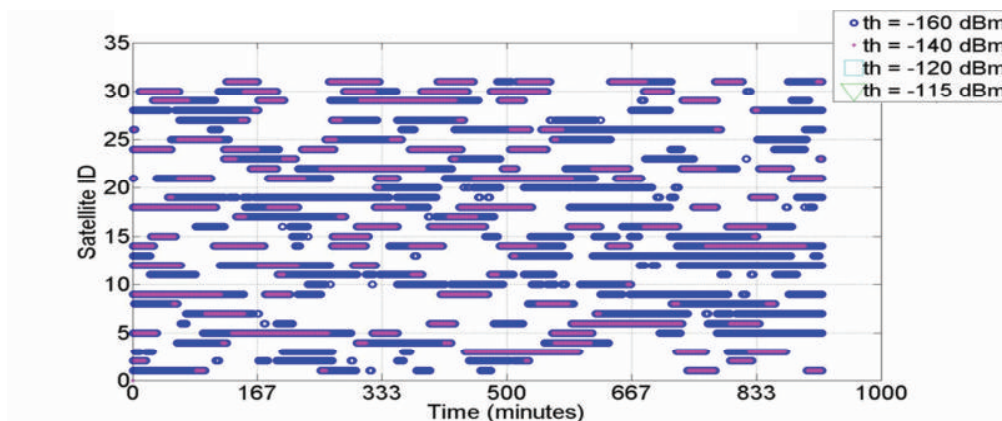


Fig. 9: GPS Satellites Availability for a MEO receiver at 25000 km.

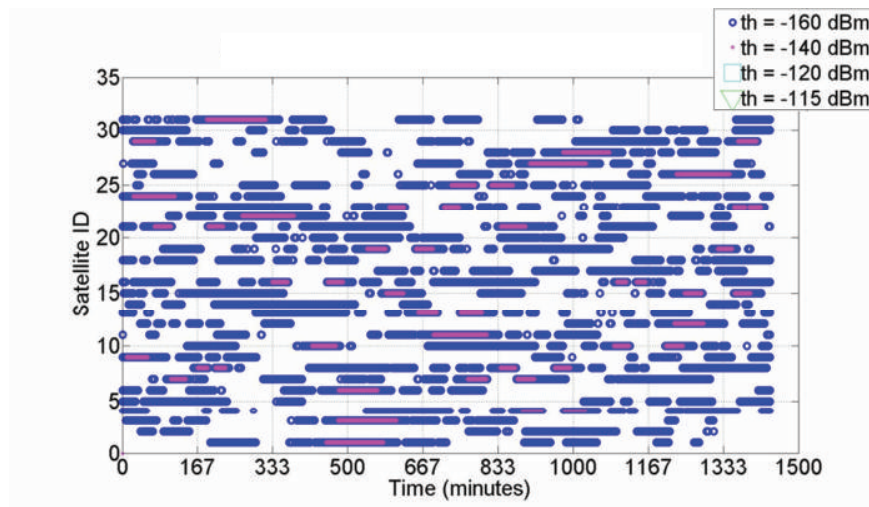


Fig. 10: GPS Satellites Availability for a GEO receiver.

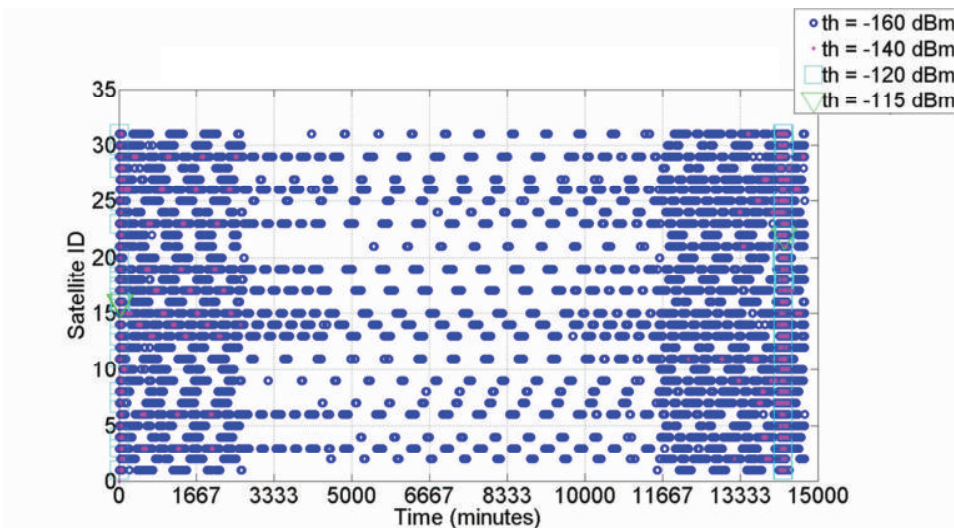


Fig. 11: GPS Satellites Availability for the defined HEO.

Time Offset

In our investigations, we did not assume a specific GNSS receiver as our aim was looking into the GNSS signals characteristics. For this reason, although the total time offset is a sum of the contributions mentioned in section II.VI, only the atmosphere delay is considered. The Table 3 contains the range of *ionospheric delays* in meters for the considered orbits.

	Ionospheric Delay (m)
MEO 1	Mean = 23.66 Max = 70.75
MEO 2	Mean = 22.30 Max = 22.34
GEO	Mean = 22.32 Max = 22.33
HEO	Mean = 22.31 Max = 22.33

Table 3: Mean and max ionospheric delay for the four orbital cases.

The results in table 3 show that the *ionosphere delay* has an approximately constant value of 22 m for MEO 2, GEO and HEO, while for MEO 1 the mean value is about 24 m with a higher maximum value of about 71 m. All the orbits here considered are equatorial, except the HEO that has an inclination of just 5° and the highest TEC occurs in the equatorial region [14]: for this reason for higher orbital inclinations, the *ionospheric delay* could be lower but not higher.

The simulations show that the *tropospheric delay* is 0 m for the strongest signals in MEO 1, MEO 2 and GEO, because they do not cross this atmosphere layer. For the considered HEO, the *troposphere delay* has a maximum of $2.7 \cdot 10^{-8}$ m just in the first low altitude part of the orbit and then it is negligible as expected.

Dilution Of Precision (DOP)

Several geometry factors can be defined to relate the various components (e.g. horizontal, vertical, etc.) of the navigation solution. In this analysis the *Geometric Dilution Of Precision (GDOP)* is considered. In order to obtain a general result, the *GDOP* is not calculated by considering the available satellites, but the visible ones. In particular, the Spirent simulator is configured to select the four satellites with the highest signal-level at the receiver position. For this reason, for a given receiver with a low power threshold, the *GDOP* could be higher than the values reported here. In fact in [1], the *GDOP* in GEO has some higher peaks that are not present here. Figures 12, 13, 14 and 15 show the *GDOP* curve trend for a GPS receiver travelling in the four different considered orbits. For MEO 1, the mean is 1.76, for MEO 2 it is 4.75, for GEO it is 10.35 and for the HEO the value changes from a minimum of 1.59 at the perigee to a maximum of 1270 at the apogee.

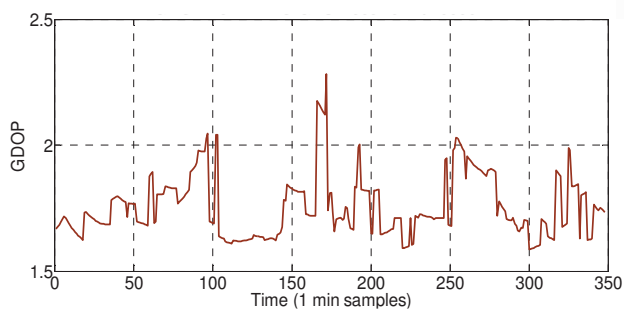


Fig. 12: GDOP for a MEO GPS receiver at 10000 km.

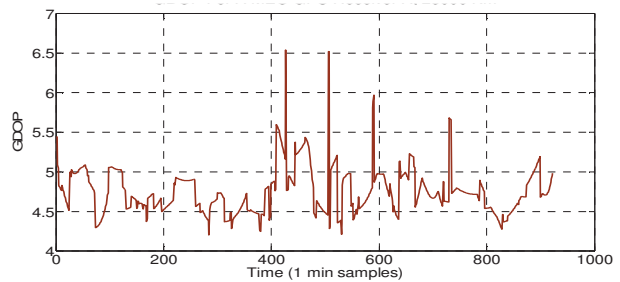


Fig. 13: GDOP for a MEO GPS receiver at 25000 km.

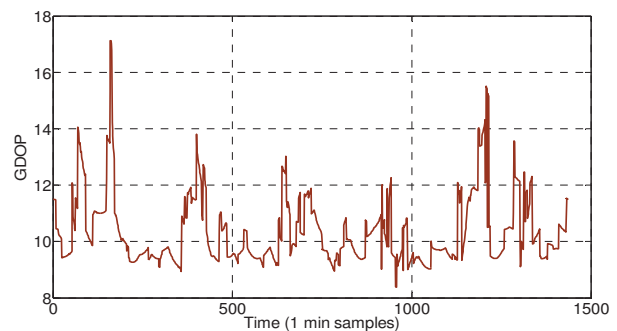


Fig. 14: GDOP for a GPS receiver in GEO.

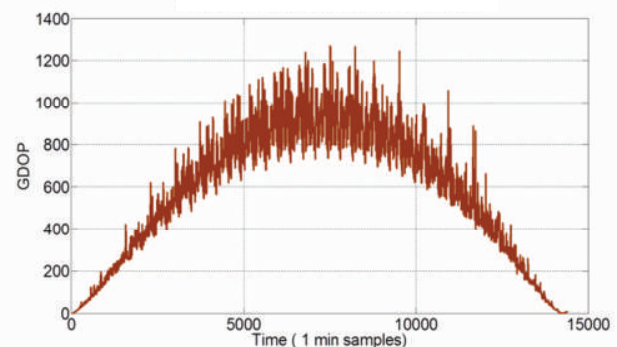


Fig. 15: GDOP for a GPS receiver in the defined HEO.

III.II GPS-Galileo-combined Constellation

Because of the increased number of GNSS satellites, we expect that the GPS-Galileo-combined constellation will provide an improved coverage, a larger availability, and an improved GDOP. Moreover, since the Galileo satellites fly at a slightly higher altitude and their transmitted power is slightly higher than GPS satellites, we also expect a higher received power. These expectations are confirmed by the following reported simulations.

Received Power

Table 4 presents for the GPS-Galileo-combined constellation the same information as was reported in

table 2 for the GPS-only constellation. Comparing both tables and as expected, there is a considerable gain of about 10 dB for the underlined power thresholds and a significant increase for the other recorded values of received power. Similar plots as reported in figures 4, 5, 6, and 7 for the probability that the received power from one or more satellites is higher than a power threshold are obtained for the GPS-Galileo-combined constellation, respectively shown in figures 16, 17, 18 and 19. Table 5 presents the received power values corresponding to 67% and 95% probability, for 1, 2, 3, 4, 5 simultaneously captured satellites, of the GPS and GPS-Galileo-combined constellations, for comparison.

	1 GPS-Galileo satellite	2 GPS-Galileo satellites	3 GPS-Galileo satellites	4 GPS-Galileo satellites
MEO 1				
$P_{r,max}$ (dBm)	-103.9	-110.6	-114.2	-114.2
$P_{r,min}$ (dBm)	-117.6	-117.9	-118.0	<u>-120.8</u>
MEO 2				
$P_{r,max}$ (dBm)	-109.8	-116.9	-119.2	-120.6
$P_{r,min}$ (dBm)	-129.4	-130.0	-130.7	<u>-131.6</u>
GEO				
$P_{r,max}$ (dBm)	-118.4	-118.4	-122.1	-122.2
$P_{r,min}$ (dBm)	-131.8	-131.9	-132.0	<u>-134.6</u>
HEO				
$P_{r,max}$ (dBm)	-99.72	-108.0	-108.2	-109.0
$P_{r,min}$ (dBm)	-149.0	-153.4	-153.7	<u>-153.8</u>

Table 4: Received power levels ranges, for the four orbital cases. The underlined minimum power $P_{r,min}$ of the ranges can be considered the minimum power threshold $P_{r,th}$ to capture four GPS-Galileo satellites simultaneously.

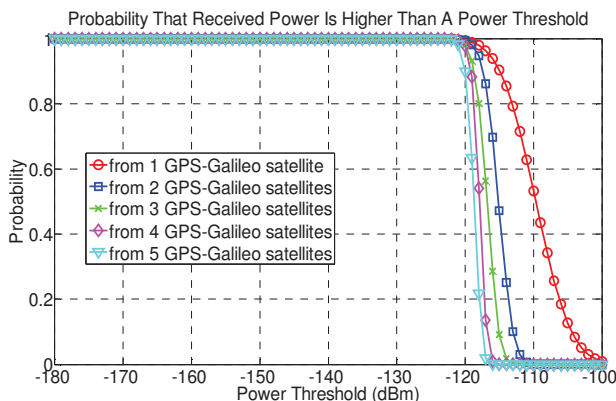


Fig. 16: probability that a received power in MEO 1 is higher than a power threshold for 1 and for 2, 3, 4, 5 GPS-Galileo satellites.

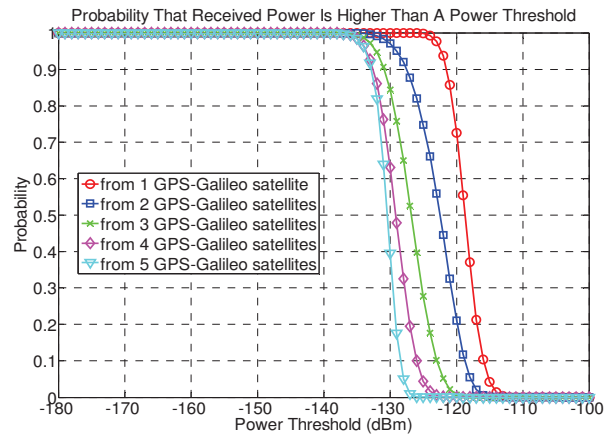


Fig. 17: probability that a received power in MEO 2 is higher than a power threshold for 1 and for 2, 3, 4, 5 GPS-Galileo satellites

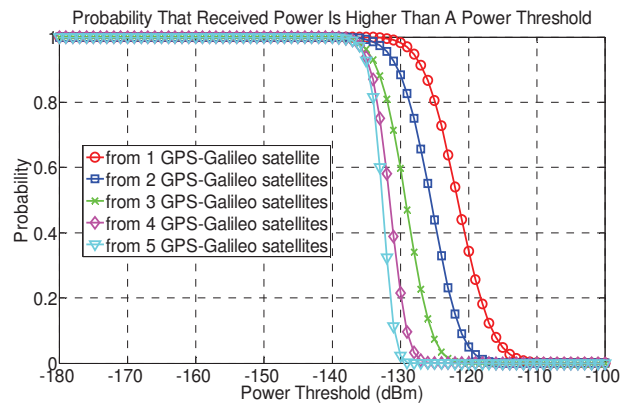


Fig. 18: probability that a received power in GEO is higher than a power threshold for 1 and for 2, 3, 4, 5 GPS-Galileo satellites

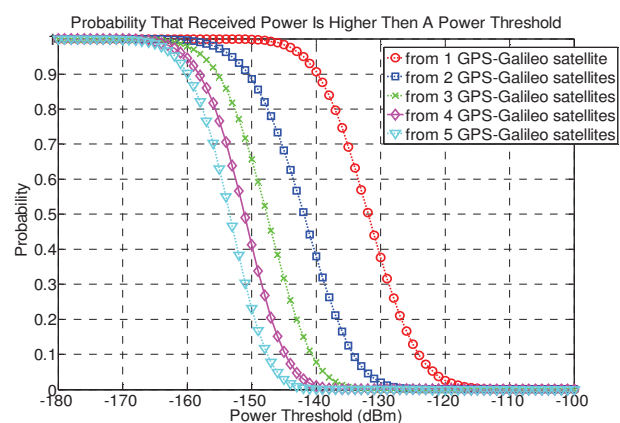


Fig. 19: probability that a received power in HEO is higher than a power threshold for 1 and for 2, 3, 4, 5 GPS-Galileo satellites.

Number of satellites simultaneously captured	Percentile	Received power (dBm) MEO 1	Received power (dBm) MEO 2	Received power (dBm) GEO	Received power (dBm) HEO
1 GPS	67 %	-119	-129	-132	-148
	95 %	-123	-133	-138	-158
2 GPS	67 %	-124	-135	-138	-157
	95 %	-127	-141	-143	-166
3 GPS	67 %	-126	-139	-141	-161
	95 %	-128	-144	-146	-168
4 GPS	67 %	-127	-141	-143	-164
	95 %	-129	-145	-146	-170
5 GPS	67 %	-128	-141	-144	-165
	95 %	-129	-145	-147	-171
1 GPS-Galileo	67 %	-111	-120	-123	-135
	95 %	-117	-122	-128	-142
2 GPS-Galileo	67 %	-116	-124	-127	-145
	95 %	-118	-129	-132	-153
3 GPS-Galileo	67 %	-117	-128	-131	-150
	95 %	-119	-132	-135	-158
4 GPS-Galileo	67 %	-118	-130	-133	-153
	95 %	-120	-134	-135	-160
5 GPS-Galileo	67 %	-119	-131	-133	-156
	95 %	-121	-134	-135	-162

Table 5: Received power values corresponding to 67% and 95% probability, for 1, 2, 3, 4, 5 simultaneously captured satellites of the GPS and GPS-Galileo-combined constellations.

Visibility and Availability

The satellite availability plots of figures 8, 9, 10 and 11 are reproduced for the case of the Galileo constellation, in figures 20, 21, 22 and 23 respectively. Obviously, by analysing figures 8 and 20 for MEO 1, figures 9 and 21 for MEO 2, figures 10 and 22 for GEO and figures 11 and 23 for the defined HEO, it is clear that the Galileo constellation offers a much better satellites availability than GPS: for a selected power threshold more GNSS satellites are available simultaneously and thanks to the higher transmitted power, in several orbital positions there is availability even where there was a GPS outage. Of course, combining both constellations will provide even further improvements.

In order to provide quantitatively the *availability* time trend of the GNSS satellite s_i , Table 6 reports:

- the average *Time Above the power Threshold* $power_{th}$ ($TAT_{power_{th}}$) during one full orbit.
- the minimum, the maximum and the average *Time of Continuity Above the power Threshold* $power_{th}$ ($TCAT_{power_{th}}$) during one full orbit.

The values in table 6 also confirm the minimum power levels of table 2 and 4.

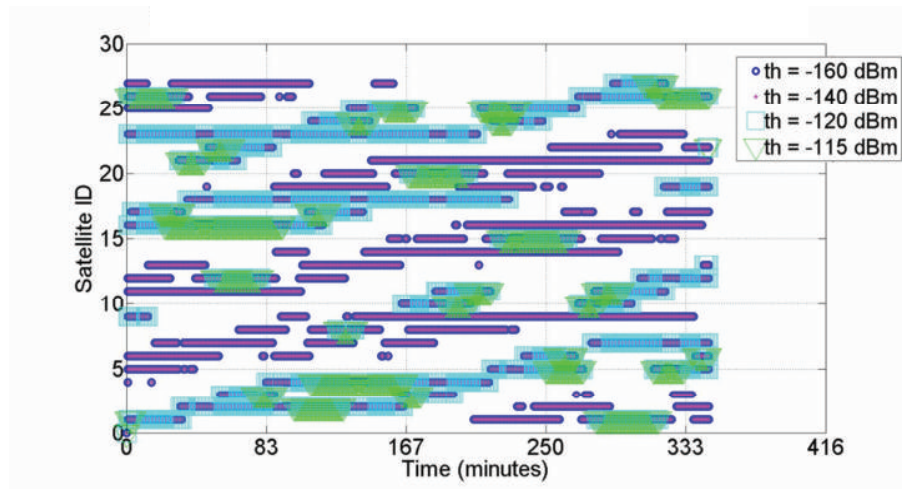


Fig. 20: Galileo Satellites Availability for a MEO receiver at 10000 km.

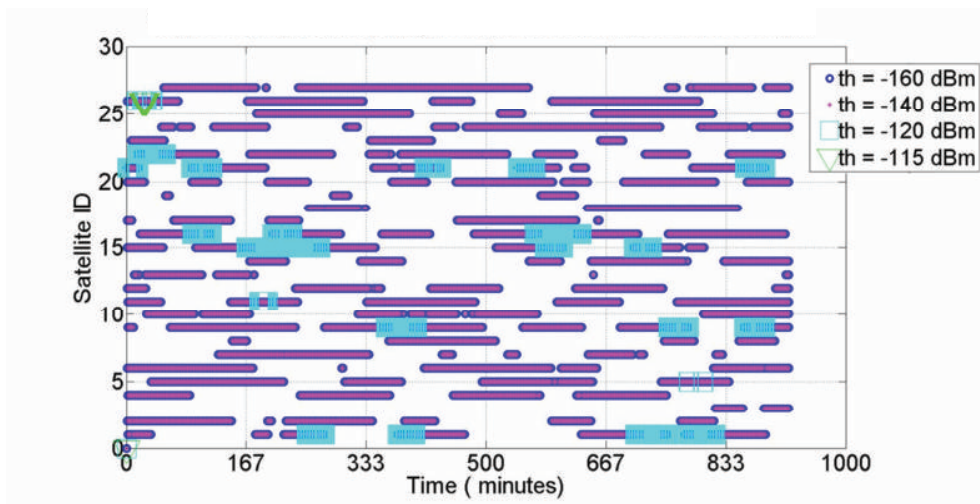


Fig. 21: Galileo Satellites Availability for a MEO receiver at 25000 km.

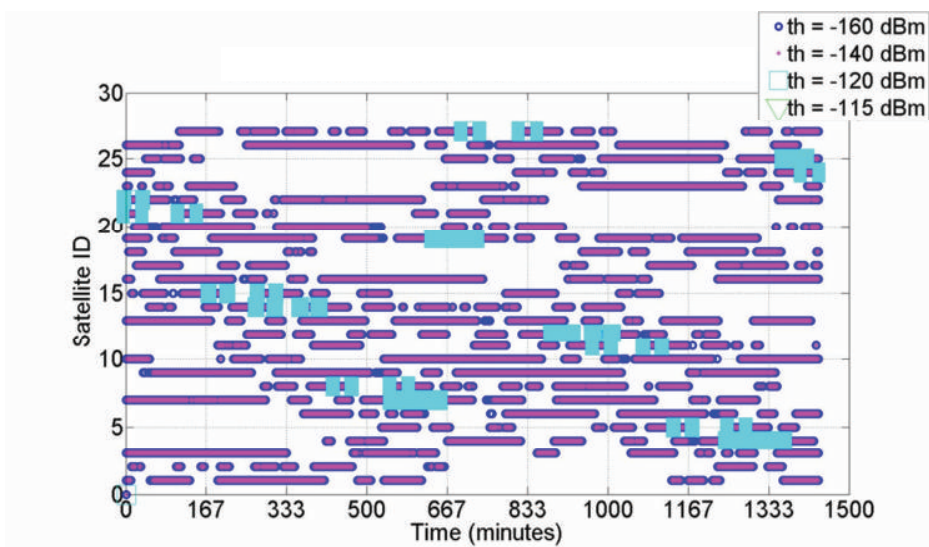


Fig. 22: Galileo Satellites Availability for a receiver in GEO.

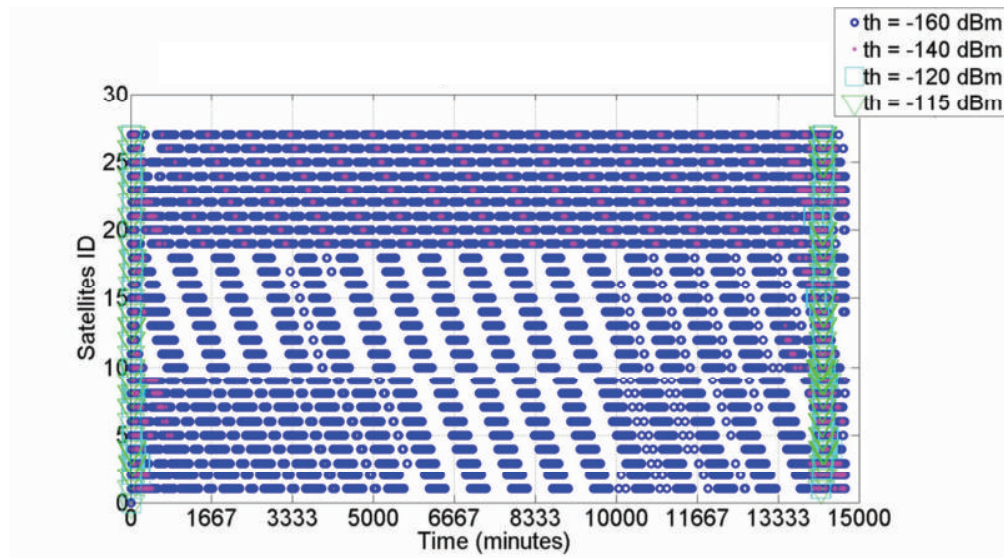


Fig. 23: Galileo Satellites Availability for a receiver in the defined HEO.

Time expressed in <i>minutes</i>	MEO 1 GPS	MEO 1 Galileo	MEO 2 GPS	MEO 2 Galileo	GEO GPS	GEO Galileo	HEO GPS	HEO Galileo
Orbital period in <i>minutes</i>	347.4		921.6		1436.0		14682.0	
$TAT_{160 \text{ dBm}}$	133.58	153.70	355.29	408.37	552.32	636.00	2755.5	6500.7
Average $TCAT_{160 \text{ dBm}}$	33.20	55.27	54.96	68.88	40.67	52.46	31.69	68.32
Min $TCAT_{160 \text{ dBm}}$	2	2	2	2	2	2	2	2
Max $TCAT_{160 \text{ dBm}}$	119	2	265	321	313	344	212	353
$TAT_{140 \text{ dBm}}$	133.58	153.70	152.48	408.37	88.77	628.44	236.68	921.70
Average $TCAT_{140 \text{ dBm}}$	33.20	55.27	35.62	68.88	58.91	55.91	20.66	35.79
Min $TCAT_{140 \text{ dBm}}$	2	2	3	2	24	2	2	2
Max $TCAT_{140 \text{ dBm}}$	119	2	153	321	135	344	112	209
$TAT_{120 \text{ dBm}}$	3.45	71.48	0	27.22	0	32.03	12.87	62.56
Average $TCAT_{120 \text{ dBm}}$	3.45	55.41	0	9.38	0	21.20	7.42	16.63
Min $TCAT_{120 \text{ dBm}}$	19	2	0	2	0	26	2	2
Max $TCAT_{120 \text{ dBm}}$	32	209	0	105	0	95	34	96
$TAT_{115 \text{ dBm}}$	1.81	24.85	0	0.26	0	0	0.58	34.41
Average $TCAT_{115 \text{ dBm}}$	1.81	18.11	0	0.26	0	0	0.42	12.95
Min $TCAT_{115 \text{ dBm}}$	12	2	0	7	0	0	3	2
Max $TCAT_{115 \text{ dBm}}$	23	71	0	7	0	0	8	56

Table 6: it contains the average *Time Above the power Threshold* $power_{th}$ ($TAT_{power_{th}}$) during one full orbit, the minimum, the maximum and the average *Time of Continuity Above the power Threshold* $power_{th}$ ($TCAT_{power_{th}}$) during one full orbit, for the four orbital cases and for the GPS and Galileo only constellations.

Dilution Of Precision (DOP)

By increasing the number of available satellites, the DOP will also be improved. In fact, for the four orbital cases, respectively figure 24, 25, 26 and 27 show the GDOP curve trend for the GPS-Galileo-combined constellations. It is calculated by considering not just four satellites, but all the visible satellites of the GPS and Galileo constellations (for this reason it can achieve values less than 1). From these simulation plots, we can see that the GDOP is much lower than the GDOP calculated only for the GPS constellation. In particular in HEO case the maximum GDOP oscillation amplitude, that corresponds to the apogee, is five times lower than that one for the GPS only constellation.

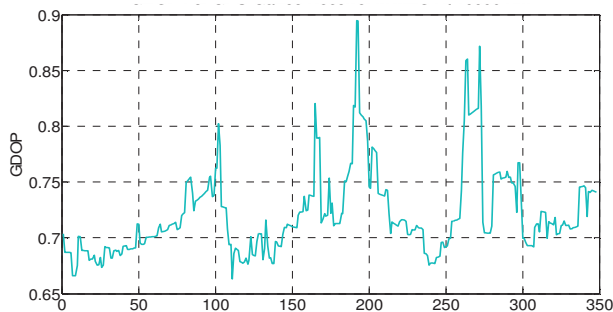


Fig. 24: GDOP for a GPS-Galileo receiver in MEO at 10000 km.

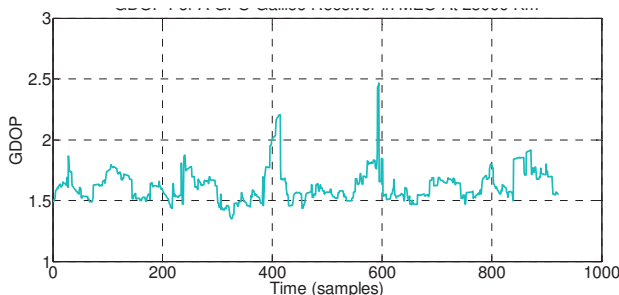


Fig. 25: GDOP for a GPS-Galileo receiver in MEO at 25000 km.

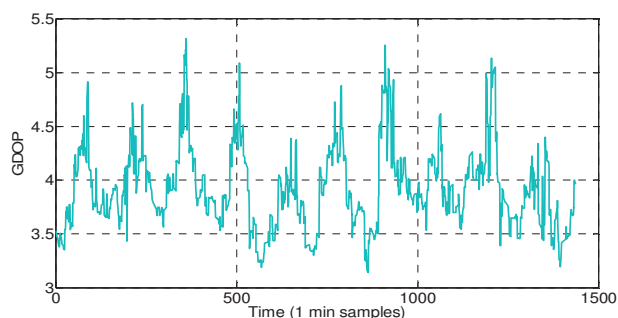


Fig. 26: GDOP for a GPS-Galileo receiver in GEO.

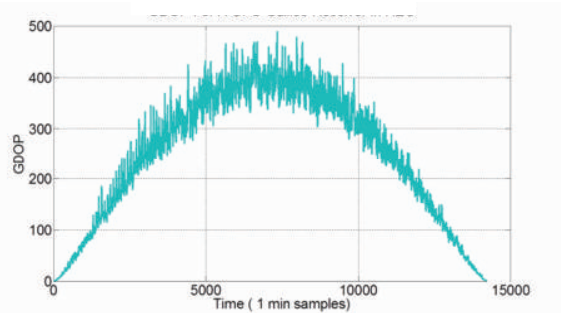


Fig. 27: GDOP for a GPS-Galileo receiver in the defined HEO.

Doppler Shifts And Doppler Rates

Because of the very high relative velocity between the GNSS satellites and the orbiting receiver, the Doppler shift is much higher than for an Earth-based receiver. On figures 28, 29, 30, and 31, we present, for the four orbital cases considered and a full orbital period, the maximum or worst-case Doppler shift considering at any time all the visible satellites. As expected, the lower is the orbit; the higher is the maximum Doppler shift because the higher is the maximum relative velocity between receiver and transmitter. The highest value of approximately 55 kHz is in fact achieved at the beginning of the HEO when the receiver is in the lowest orbital point and accordingly it has the maximum velocity. Moreover, a considerable value of 20.5 kHz is achieved in MEO 1. In order to know how the Doppler shift changes in one full orbit for a given GNSS satellite, figures 32, 33, 34 and 35 illustrate the histogram of the Doppler rate values calculated for a receiver in the four considered orbits and the GPS and Galileo satellites.

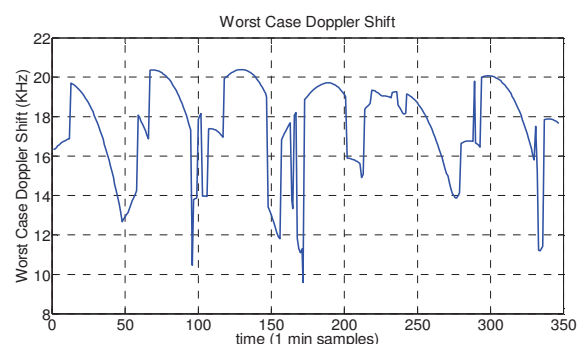


Fig. 28: Worst case Doppler shift for a GPS-Galileo receiver in MEO 1.

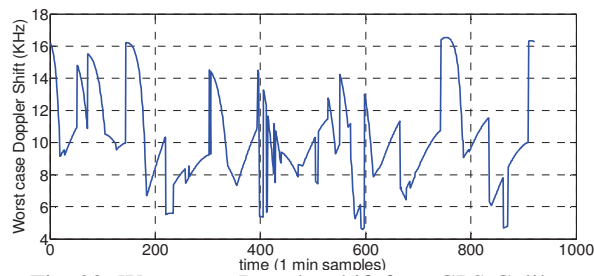


Fig. 29: Worst case Doppler shift for a GPS-Galileo receiver in MEO 2.

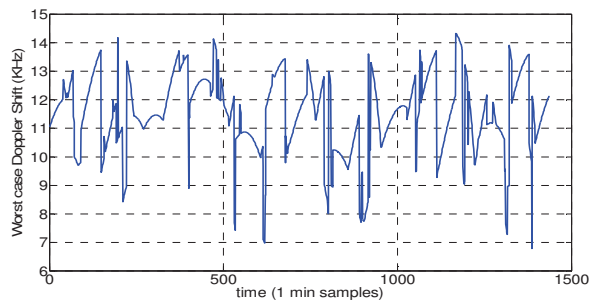


Fig. 30: Worst case Doppler shift for a GPS-Galileo receiver in GEO.

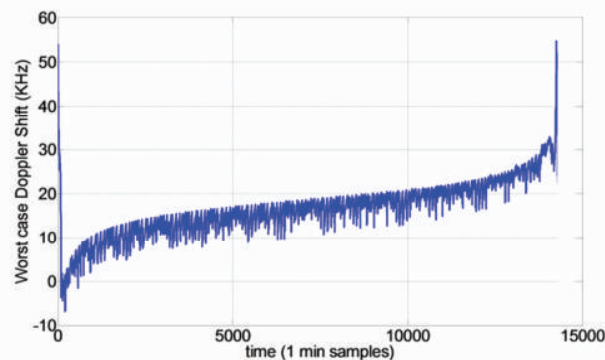


Fig. 31: Worst case Doppler shift for a GPS-Galileo receiver in the defined HEO.

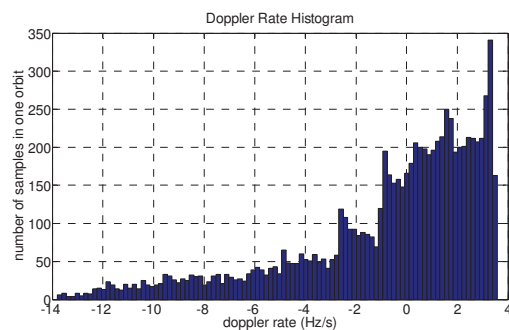


Fig. 32: Doppler rate histogram for a GPS-Galileo receiver in MEO 1.

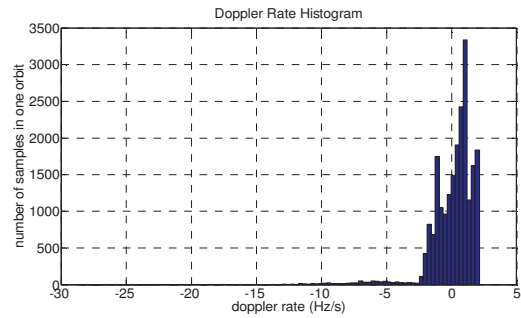


Fig. 33: Doppler rate histogram for a GPS-Galileo receiver in MEO 2.

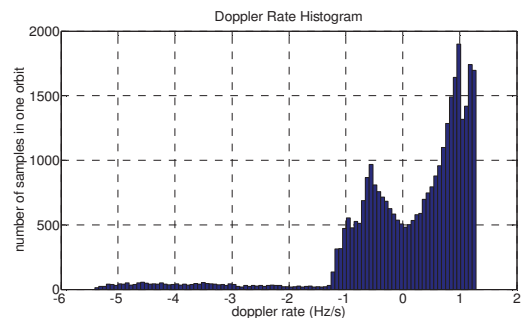


Fig. 34: Doppler rate histogram for a GPS-Galileo receiver in GEO.

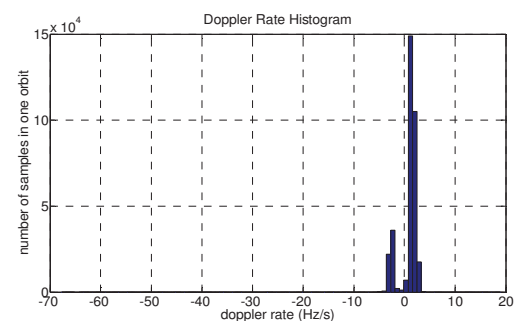


Fig. 35: Doppler rate histogram for a GPS-Galileo receiver in the defined HEO.

III.III Comparison With Other References

The presented results can be compared to the results in [1], [12] and [13]. Particularly in [1] and [12] the GEO orbital case is simulated and the power levels are expressed as carrier-to-noise ratio of the signal following analog-to-digital conversion: this is done by choosing thresholds compatible with the expected performances of the receiver in term of positioning, of precision, of implementation complexity, etc. As already described, in order to provide valid results for more than one specific receiver, in this paper just the received power at the receiver position is provided. However, as in [1], it may be possible to calculate the carrier-to-noise ratio by using the following formula:

$$C/N_0 = P_r - 10\log_{10}(kT_{sys}) - L_{nf} - L_i \quad [2]$$

Where:

C/N_0	is the carrier-to-noise ratio
P_r	is the received power in dBW
k	is the Boltzmann's constant,
T_{sys}	is the effective system noise temperature in Kelvin
L_{nf}	is the noise figure of the receiver front-end in dB
L_i	is the implementation and A/D conversion losses in dB.

If as in [1] we assume $L_{nf} = +3.5$ dB, $L_i = +2$ dB and $T_{sys} = 190$ K and considering the values in table 2 to acquire and/or track at least four GPS satellites simultaneously, the carrier-to-noise ratio in the GEO orbital case will have a minimum of 25.4 dB-Hz and a peak of 40.3 dB-Hz. The minimum of 25.4 dB-Hz is therefore not too far from the reported value of 29 dB-Hz in [1] and [12]. One reason for this difference is because in [1] the received power P_r is calculated as a sum of constant contributions, by assuming a constant gain for the transmitting antenna, while in our study, as already described, the gain is function of the antenna elevation and azimuth.

IV. CONCLUSIONS

The simulation performed using the multi-GNSS constellation simulator "Spirent GSS8000" have shown four examples of GNSS use for space applications from MEO to HEO with apogee close to the Lunar altitude. The simulations results indicate that in order to acquire and track at least four satellites simultaneously, a GPS receiver must be designed in such a way to acquire and track a -134.2 dBm signal in MEO 1, a -141.2 dBm signal in MEO 2, a -144.9 dBm signal in GEO and a -162.7 dBm in the defined HEO. Similarly, a GPS-Galileo-combined receiver will need to acquire and track a -120.8 dBm signal in MEO 1, a -131.6 dBm signal in MEO 2, a -134.6 dBm signal in GEO and a -153.8 dBm in the defined HEO. Furthermore the receiver should be able to acquire/track signals affected by a Doppler shift of approximately 21 kHz in MEO 1, 17 kHz in MEO 2, 15 kHz in GEO and 55 kHz in the low altitude part of the defined HEO. As expected, the MEO 1 is the orbit that shows the highest Doppler rate absolute value of almost 14 Hz/s for a not negligible number of orbit samples. The atmosphere delay

simulations have shown a mean delay of about 22 meters for MEO 2, GEO and the defined HEO and a mean of about 24 meters for MEO 1. The GDOP analysis instead, has proved the relative degradation of geometry as the altitude increases, as expected. Finally, simulation results have shown the significant GNSS performance improvements achievable by using a multi constellation receiver.

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